

Table 1: Features of AWS WAR and their benefits

Feature	Description	Benefits
Comprehensive framework	AWS Well-Architected Reviews provide a structured framework consisting of six pillars: operational excellence, security, reliability, performance efficiency, cost optimisation and sustainability.	This helps in systematically evaluating and optimising cloud architectures to meet the requirements of modern software development, ensuring that applications are built with best practices and adhere to industry standards.
Guidance and best practices	Offers guidance and best practices for designing, deploying, and optimising cloud-based architectures.	Assists in making informed decisions during cloud-native migration, modernisation, and transformation initiatives, enabling organisations to leverage proven strategies and approaches for building scalable and resilient cloud applications.
Identifying design flaws	Helps in identifying design flaws and architectural bottlenecks early in the development life cycle.	Facilitates proactive identification and resolution of potential issues, minimising the risk of costly rework and ensuring that cloud-native applications are designed to meet performance, security, and reliability requirements.
Continuous improvement	Emphasises the importance of continuous improvement and iterative refinement of cloud architectures.	Encourages organisations to adopt a culture of continuous improvement, fostering innovation and agility in software development processes, and enabling teams to evolve and adapt their cloud architectures over time.
Cost optimisation	Provides strategies for optimising costs without sacrificing performance or reliability.	Helps in maximising cost efficiency in cloud-native migration and modernisation projects by identifying opportunities for cost optimisation, right-sizing resources, and leveraging cost-effective architectural patterns and AWS services.
Security best practices	Addresses security best practices, including data protection, identity and access management, and compliance.	Ensures that cloud-native applications are built with robust security controls and measures in place, protecting sensitive data and mitigating risks associated with cyber threats and compliance requirements.
Reliability and resilience	Focuses on ensuring that systems can recover from failures and meet business requirements for availability.	Helps in designing cloud-native applications with built-in fault tolerance, disaster recovery, and high availability capabilities, ensuring continuous operation and minimising downtime in the event of failures or disruptions.
Performance optimisation	Aims to optimise resource utilisation and maximise performance to meet business requirements.	Assists in improving the performance and scalability of cloud-native applications by optimising resource allocation, implementing caching and load balancing techniques, and monitoring performance metrics to identify bottlenecks and areas for improvement.
Tailored recommendations	Offers tailored recommendations based on workload requirements, usage patterns, and business objectives.	Provides actionable insights and recommendations for optimising cloud architectures based on specific use cases and requirements, helping organisations tailor their strategies and approaches to meet their unique business needs.

code and cost benefits as needed.

- **Anticipate failure** – Improve your validation procedure to break the code so that you can fix things early.
- **Learn from all operational failures** – Avoid repeating mistakes and improve your visionary thought to

build quality code.

Enabling efficient cloud migration

Let's consider a scenario where a retail store or banking organisation is planning to migrate its on-premises

data centre to a public cloud like AWS. The company has a complex architecture with multiple applications, integration services, databases and dependencies. Here's how AWS WAR can help improve the efficiency of this cloud migration journey.